REPUBLIC OF THE PHILIPPINES
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
BIÑAN CAMPUS

BAGONG PILIPINAS

Embedded Systems

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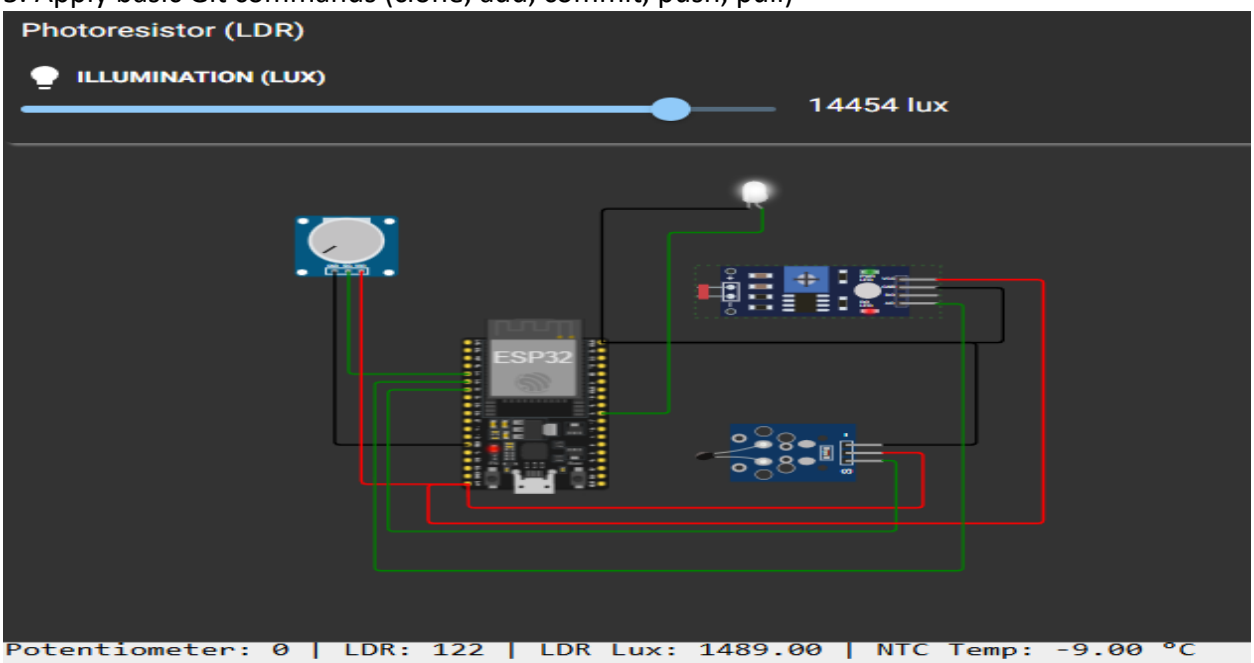
Finals Period **AC**	**Activity#2: ESP32 ADC + PWM with GitHub Version Control** **Members: Francis Oliveros, John Allen Hidalgo, Kristine May Martillano, Paolo Lazareto, Chayzie Tamayo** **Group Number: 7** **Section: BSCPE Ladderized**	**Grade**

I - Objectives:

1. Read analog input using ESP32 (ADC)
2. Control LED brightness using PWM
3. Use sensors (Potentiometer, LDR, and NTC)
4. Upload and manage project using GitHub
5. Apply basic Git commands (clone, add, commit, push, pull)




```
Potentiometer: 0 | LDR: 122 | LDR Lux: 1489.00 | NTC Temp: -9.00 °C
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```

```
// Pins
const int potPin = 34;
const int ldrPin = 35;
const int ledPin = 25;

// PWM
const int pwmFreq = 5000;
const int pwmResolution = 8;

void setup() {
  Serial.begin(115200);
  ledcAttach(ledPin, pwmFreq, pwmResolution);
}

void loop() {

  // ========================
  // A1: Potentiometer → LED Brightness
  // ========================
  int potValue = analogRead(potPin);
  int brightness = map(potValue, 0, 4095, 0, 255);

  // ========================
  // A2: LDR → Light Detection
  // ========================
  int ldrValue = analogRead(ldrPin);

  // Convert to "lux" (scaled for display only)
  float lux = map(ldrValue, 0, 4095, 0, 50000);

  // Auto LED control
  if (ldrValue > 2000) {
    brightness = 0; // bright → OFF
  }

  ledcWrite(ledPin, brightness);

  // ========================
  // A3: "Temperature" controlled by LDR
  // ========================
  // Map LDR to temperature range (adjustable)
```

```
  float temperatureC = map(ldrValue, 0, 4095, -10, 50);

  // ========================
  // SERIAL OUTPUT (your format)
  // ========================
  Serial.print("Potentiometer: ");
  Serial.print(potValue);

  Serial.print(" | LDR: ");
  Serial.print(ldrValue);

  Serial.print(" | LDR Lux: ");
  Serial.print(lux);

  Serial.print(" | NTC Temp: ");
  Serial.print(temperatureC);
  Serial.println(" °C");

  delay(500);
}
```

Conclusion:

This project demonstrates how an ESP32 can integrate multiple sensors to create an interactive and responsive system.

- The **potentiometer (A1)** provides manual control, allowing the user to adjust LED brightness smoothly.
- The **LDR (A2)** enables automatic behavior, turning the LED off in bright conditions and on in darker environments, improving energy efficiency.
- The **temperature function (A3)** shows how sensor data can be interpreted and displayed. In the modified version, the LDR is also used to simulate temperature, illustrating how one input can control multiple outputs for demonstration purposes.

Overall, the system highlights key embedded system concepts: **analog input reading, PWM output control, sensor-based decision making, and data monitoring via serial communication**. It also shows the importance of proper calibration and wiring to ensure accurate and stable readings.

This setup can be extended further into real-world applications such as **smart lighting systems, environmental monitoring, or home automation projects**.